A comment on 'Growth and Inequality in Public Good Provision': Testing the Robustness and Generalizability of Dynamic Public Good Games

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Abstract

I revisit the dynamic public goods game by Gächter et al. (2017) that captures temporal interdependencies in cooperation. I first reproduce parts of the original analysis and find only minor and inconsequential discrepancies. Using both student and a more representative sample, I then replicate findings about growth and inequality in public good provision. Finally, I examine whether the robustness of results also translates into generalizability. Specifically, I test whether behavior in this experimentally-induced social dilemma predicts real climate action through voluntary carbon offsetting. Despite the game's enhanced ecological validity through temporal interdependencies, I find no correlation between game behavior and climate action in either sample. This suggestive evidence indicates that laboratory paradigms, even when incorporating key real-world features, may better serve to isolate specific behavioral mechanisms than predict field behavior.

Keywords: Public goods, Dynamic games, Reproducibility, Generalizability, Online experiment, Climate change mitigation

1. Introduction

Current actions shape tomorrow's possibilities. Many decision-making contexts exhibit temporal interdependencies, where today's choices not only affect immediate outcomes but also determine future decision spaces. Consider environmental policy: implementing eco-friendly measures early has two advantages: reducing immediate carbon dioxide emissions while also making future climate goals, such as the Paris Agreement targets, more attainable. Whereas delaying such policies doesn't necessarily preclude reaching these targets, it demands significantly greater future efforts compared to immediate action (Hänsel et al., 2022). Thus,

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when pursuing specific goals, present actions (or their absence) influence both intermediate outcomes and the range of viable future pathways.

Public good (or public bad) games provide a versatile framework for exploring cooperative behavior in various contexts. By simulating scenarios where individual and collective interests conflict, these experimental paradigms shed light on the dynamics of collective action. Hence, they can be applied to better understand challenges in areas such as healthcare, infrastructure, education, and environmental management. In such applications, public good games can be used to isolate key mechanisms of group cooperation—such as punishment (Fehr and Gächter, 2000), communication (Tavoni et al., 2011), reputation (Milinski et al., 2002), and thresholds (Cadsby and Maynes, 1998)—to guide the design and implementation of policies addressing real-world social dilemmas.

However, many real-world cooperation problems—particularly in environmental policy—involve temporal interdependencies that traditional public good games fail to capture. While these games are frequently used to inform climate policies (e.g. Milinski et al., 2006; Hauser et al., 2014; Brick and Visser, 2015; Vicens et al., 2018; Calzolari et al., 2018; Cook et al., 2019), they have a key limitation: participants are given the same choices in each round, regardless of previous decisions. As a result, current decisions have no bearing on future action spaces. This limitation affects the ecological validity (meaning the extent to which experimental findings resemble real-world dynamics) for studying climate change, where carbon dioxide emissions accumulate and persist in the atmosphere for over a millennium (Inman, 2008; Calzolari et al., 2018). Traditional public good games reset each period, allowing participants to disregard the cumulative impact of their past decisions.

Gächter, Mengel, Tsakas & Vostroknutov (2017, hereafter, GMTV) and Stefan Große (unpublished) address this limitation through a *dynamic* public goods game (dPGG). By defining endowments as earnings from previous periods, they create direct links between current actions and future possibilities: higher (lower) current earnings expand (contract) future contribution options. Crucially, while introducing these temporal interdependencies, the dPGG maintains the core predictions of static games (free-riding and public good under-provision) making it an interesting paradigm for studying social dilemmas (see also Battaglini et al., 2016; Rockenbach and Wolff, 2017; Eichenseer and Moser, 2019)

GMTV found that, even without punishment, the dynamic setting generated increasing absolute contributions over time, though contributions as a share of endowments declined from 55% to 35%. Groups achieved substantial growth but with considerable heterogeneity–the richest group earned more than 10 times what the poorest group earned after 10 periods. Inequality increased over time but remained moderate (Gini $\approx$ 0.2), with more successful groups showing lower inequality. Notably, the strong correlation between early and final wealth ($\rho = 0.48$) suggests path dependence, where initial decisions significantly shaped groups'

trajectory.

To better evaluate these results, I first conducted a "recreation reproduction" (Brodeur et al., 2024b, p. 9) one of GMTV's treatments. Using their raw data but developing new code in R, I successfully reproduced the qualitative results of the relevant treatment, finding only minor numerical differences despite some discrepancies in the authors' code. I then conducted a conceptual replication by adapting GMTV's laboratory experiment to an online setting during the COVID-19 pandemic. To test the robustness of their findings, I examined both student and a more representative samples, finding behavior patterns consistent with those reported in the original paper.

Whereas robustness across populations and over time establishes the reliability of the experimental findings, their generalizability to real-world behavior remains an open question. Goeschl et al. (2020) demonstrate that static public goods games do not effectively predict real-world climate action, with generalizability depending on ecological validity, that is, how closely the game's structure mirrors climate change mitigation contexts. Given that GMTV's dynamic setting incorporates temporal interdependencies—a key feature of climate change—one might expect it to better inform public policy. To test this hypothesis, I complemented the conceptual replication with observations of participants' voluntary climate actions (VCA). In their VCA, participants acted as dictators, choosing how to divide their budget between themselves and organizations focused on CO2 emissions reduction (Eckel and Grossman, 1996; Carpenter et al., 2008). This approach parallels Goeschl et al. (2020)'s methodology, enabling analysis of how behavior in the abstract game translates to real-world action across. Contrary to expectations, I do not find evidence in favor of generalizability.

After reporting the methods in Section 2, this paper is organized along these findings: Section 3 first reports the confirmatory reproduction in Section 3.1 before documenting the conceptual replication and discussing the generalizability in Section 3.2 and Section 3.3, respectively. Section 4 concludes.

2. Methodology

Whereas GMTV also designed a treatment arm in which participants were able to subtract tokens from other participants, I only focus on a 10-period game without punishment. In what follows, I'll describe the design of the dynamic game in more detail. I also describe the VCA, which differs slightly from GMTV's design. Subsequently, I describe the sample and procedure before I turn to the results.

2.1. Experimental Design

The experimental design builds on the workhorse model Zelmer (2003, p. 301) describes in her meta-analysis, where

> *subjects are divided into groups and play the same game for a finite number of periods. In each period, every subject is endowed with an income […] The subject must then divide this income between a contribution to a private account […] that yields a constant return to themselves only and a contribution to a public account […] where consumption benefits accrue to all group members. At the end of each period, subjects typically learn the aggregate contribution to the public good by all members of their group and their earnings for the period.*

GMTV introduced dynamics to this standard setup through their NOPUNISH and PUNISH treatments, each running for either 10 or 15 periods. In their NOPUNISH treatments, participants played the standard public goods game with one key modification: rather than receiving fresh endowments each period, participants' endowments were determined by their accumulated wealth from previous periods. Specifically, participants receive one endowment only at the beginning of the first period. A participant's endowment in the second period is the wealth she accumulated in the first period. A participant's endowment in the third period is the wealth she accumulated in the first two periods. And so on. Hence, a decision in one period has consequences on future endowments and, ultimately, growth paths. For this reason, the game is described as a *dynamic* public goods game. GMTV's PUNISH treatments added the ability for participants to reduce others' earnings at a cost.

I focus on replicating their NOPUNISH10 treatment, maintaining identical parameters: I ran sessions with groups of four ($i \in I = \{1, 2, 3, 4\}$), an initial endowment of $N_i^1 = 20$ tokens, $T = 10$ periods, a private account with a return of 1 and a group account with a return of 1.5 ($\Rightarrow$ MPCR$\equiv \frac{1.5}{4}$). As in GMTV's experiments, a token's nominal worth was 0.05 Euros. With i's contribution in period t being c_i^t, the model looks as follows:

$$N_i^{t+1} = N_i^t - c_i^t + \frac{1.5}{4} \sum_{j=1}^{4} c_j^t$$

2.2. Voluntary Climate Action (VCA)

In the same stroke as GMTV I employ a real giving task after the abstract game. Specifically, I employed a (charitable) dictator game where each participant is a dictator dividing her budget between herself and some organization. Whereas GMTV chose *Doctors without Borders* as a receiver, I chose another organization in another context: like Goeschl et al. (2020), I employed a VCA, where participants could donate any amount of their earnings to offset carbon dioxide (that is, retire emission permits from the EU ETS).[2]

[2]Importantly, Goeschl et al. (2020) made the VCA decision with a fresh endowment *before* they played the abstract game. I deviate from their procedure to match GMTV's procedure.

To ensure that each participant had the same basic level of information about the impact of their decision, I provided some basic information about the mechanism. The information also highlighted that the mitigation came into operation on a European level. Finally, I informed the participants that the documentation of individual and aggregate contributions was to be posted immediately after the conclusion of the final session online. To avoid privacy or social image concerns, participants learned their unique and random IDs, which they needed to identify their individual contributions. The document also certified that the sum of their contributions have been used to offset 1.82 tons of carbon dioxide emissions (see https://tinyurl.com/4fnt2vrh).

2.3. Recruitment and Sample Characteristics

I recruited the participants from the so-called HamburgPanel using HROOT (Bock et al., 2014). The panel is provided by the University of Hamburg's Research Laboratory, which used a randomized last digits approach to build the panel while drawing from the population of citizens of Hamburg, Germany. Whereas this sampling approach provides a *more* representative sample than typical student populations, it's important to note that some selection bias remains: participation requires both internet access and sufficient digital literacy to engage in online experiments, and individuals must have opted into the panel initially. For clarity, I refer to this as a more representative sample throughout the paper, while acknowledging it is not a *true* general population sample.

The panel counted 1.209 non-students of which I was able to recruit only 130 participants who finished the experiment—even though I varied the weekdays and timing of the sessions (which were conducted during a nationwide lockdown under mandatory home office regime). For this reason, I also recruited students in the last session. I provide more details in Appendix D.

At the time I conducted the experiment, the more representative sample was not familiar with interactive experiments. In fact, I ran the first interactive group experiment with this sample. The students, in contrast, were used to single-player experiments. Recruiting them, I excluded those who have participated in interactive experiments such as public goods games before, to keep the two distinct subject pools comparable in terms of their experience. As a consequence, nonnaiveté is unlikely to affect the validity of the experiment (Goodman and Paolacci, 2017, p. 204).

Throughout large parts of this paper, I will compare the results of my experiment with the results of GMTV's NOPUNISH 10 Period treatments. I am thus, referring to three different samples utilized at two different points in time: the University of Maastricht's students (in late 2012), Hamburg's citizens, and the University of Hamburg's students (both in July 2021).

Figure 1 illustrates the age distributions across samples, with Hamburg citizens showing greater diversity

compared to both student groups. As detailed in Table 3 (see Appendix C), the samples differed significantly in both age (F=87.278, p<0.01) and gender composition (F=2.596, p<0.1), with Hamburg citizens being notably older (mean = 47.58, sd = 14.98) than both student samples (Maastricht: mean = 47.58, sd = NA and Hamburg: mean = 34.73, sd = 7.59).

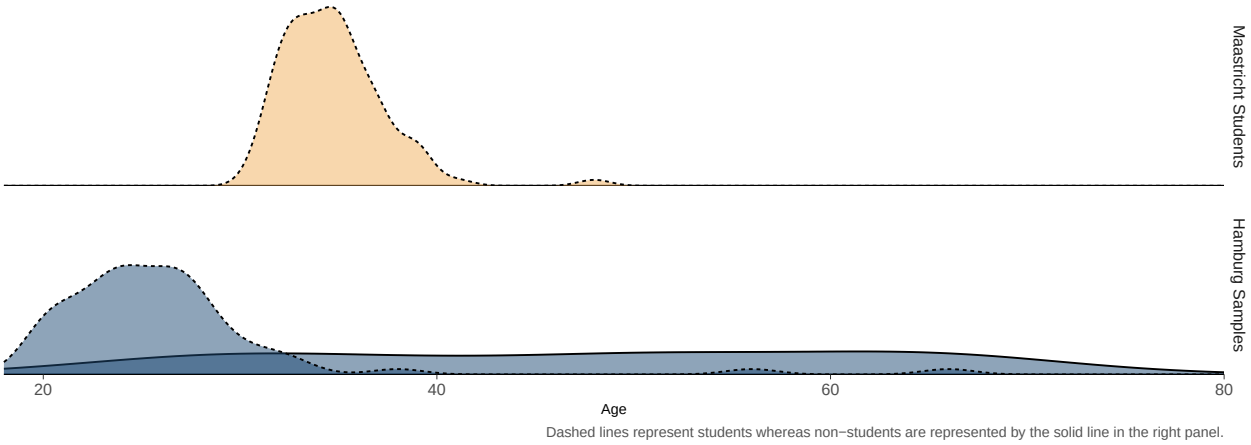


Figure 1: Age Distribution Across Samples

2.4. Procedure

Participants entered the experiment at appointed times remotely from home. They first saw a welcome screen. After agreeing to the privacy policy, they could proceed to the instructions individually. Having read these instructions, each participant has also seen a demo-screen explaining the user interface. Before proceeding, they had to answer six comprehension questions correctly to avoid confusion in later stages (Ferraro and Vossler, 2010). Subsequently, they saw a waiting screen until they could be matched with three other participants, who have answered the comprehension questions correctly. Once matched, they were exposed to the decision screen over ten periods. At the end of the last period, participants saw the results of all periods. Subsequently, they made their VCA decision, before I elicited risk preferences (Holt and Laury, 2002) and finished with GMTV's questionnaire.[3]

As the experiment was conducted online with an inexperienced sample, dropouts required special attention. Participants were grouped according to the order in which they answered the comprehension questions correctly, reducing waiting times. While waiting to form a group, participants saw a screen informing them they would not have to wait longer than 10 minutes and would receive a *"patience bonus"* of one Euro if the full waiting time elapsed. To maintain the experiment's momentum, participants had 10 minutes to make their first contribution and 4 minutes for subsequent contributions. If these time limits expired, the

[3]The risk elicitation and questionnaire were included to maintain comparability with GMTV's protocol but are not analyzed here as they fall outside this paper's focus on replication and generalizability of the main behavioral patterns.

participant was replaced by a bot making uniformly distributed random contributions between 0 and their endowment. The remaining group members were informed when such replacements occurred. These features were implemented to minimize wait times and maintain engagement for active participants.

Overall, I adhered closely to GMTV's protocol but made three necessary modifications. First, the instructions were in German and included additional information about online-specific features (such as dropouts and bots). Second, I used oTree instead of zTree (Chen et al., 2016; Fischbacher, 2007), which resulted in a different graphical user interface. Third, and most importantly, I replaced GMTV's charitable giving option (Doctors without Borders) with voluntary carbon offsetting through EU ETS permit retirement. This third modification was not intended to replicate GMTV's charitable giving analysis, but rather to follow Goeschl et al. (2020)'s methodology for examining climate action specifically. By using carbon offsets instead of general charitable giving, I could directly test whether a dynamic game that captures temporal interdependencies—a key feature of climate change—can predict real-world climate mitigation behavior.

The experiment lasted on average for around 25 minutes. The overall earnings (including the incentivized risk elicitation task) averaged 11.23 Euros (sd = 4.85). For context, the German minimum wage at the time was 9.50 Euros per hour, meaning participants earned on average more than one hour's minimum wage for a 25-minute experiment.

3. Results

This section comprises my main results and is organized along my three main contributions: Section 3.1 describes and discusses the recreation reproduction, Section 3.2 the conceptual replication, and Section 3.3 the generalizability of GTMV's results.

3.1. Recreation Reproduction

Not only replication but also its predecessor, reproduction, are fundamental to scientific progress, serving as diagnostic tools to validate and build confidence in research findings (Brodeur et al., 2024b). Current research shows that many published studies remain computationally irreproducible–meaning that even with the original data and code, other researchers cannot reproduce the reported results (Chang and Li, 2017, Stodden et al. (2018); Trisovic et al., 2022). This undermines the credibility of scientific research and hampers the accumulation of reliable knowledge.

Using the original data provided by GTMV and translating their STATA code to R, I identified three issues that required attention (see Appendix A for detailed discussion). First, their code incorrectly computed Gini coefficients when participants had identical endowments. Second, the calculation of contribution shares

used a lagged endowment measure, leading to systematic bias in these values. Third, wealth and inequality measures were defined at period beginnings rather than endings, missing important end-game effects.

To assess how these corrections affect the data, I compare outcomes before and after implementing them. The error concerning the Gini coefficient only affects cases with identical endowments, which occurred in about 13% of group × period level observations succeeding the first period, irrespective of the treatment. The *share of endowments contributed* calculation has a more systematic impact, tending to understating contribution shares. However this bias is also uniform across treatments and does not affect comparative analyses in the original paper. The same argument applies to the authors' reliance on beginning-of-period behavior.

Figure 2 visualizes both the authors' original as well as my corrected measures for wealth, contribution behavior, and inequality over time in the NOPUNISH10 Treatment. The corrections reveal systematically higher wealth accumulation and inequality levels than originally reported. Similarly, the contribution shares have been overestimated byt the authors. Despite these quantitative differences, the qualitative patterns identified in GTMV's analysis—increasing wealth and inequality coupled with declining contributions—remain robust.

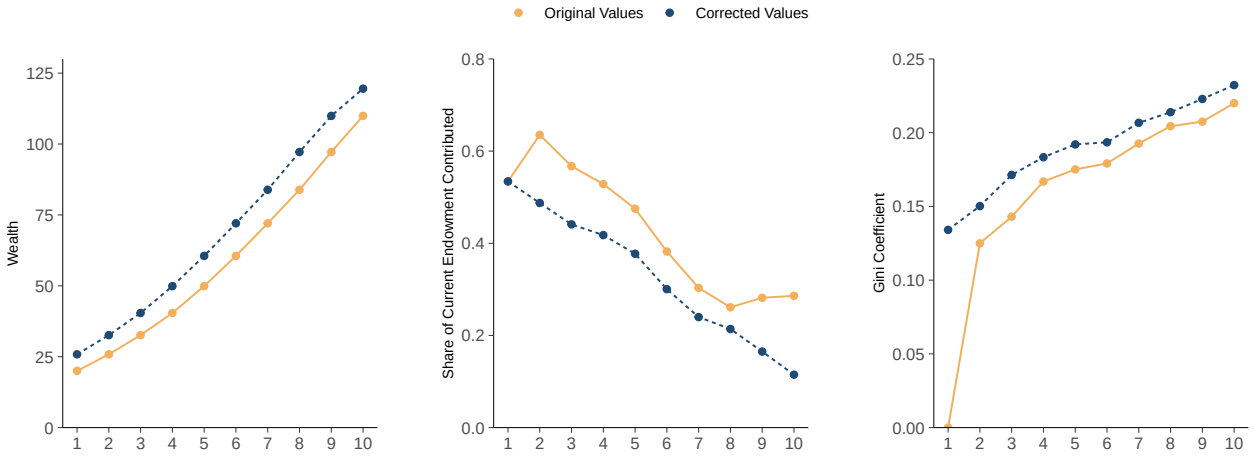


Figure 2: Comparison of Original and Corrected Values for Wealth, Inequality, and Contributions in the NOPUNISH10 Treatment

3.2. Conceptual Replication

Whereas reproduction using original data helps identify computational issues, replication studies with new data are crucial for assessing the robustness of research findings to answer the empirical question of whether findings are externally valid. Recent large-scale replication projects across various disciplines have highlighted both successes and challenges in replicating published results (Open Science Collaboration, 2015; Camerer et al., 2018; Fišar et al., 2023; Brodeur et al., 2024a). In economics specifically, replication rates are

about 60% (Camerer et al., 2016), with successful replications often showing smaller effect sizes than original studies. Moreover, methodological choices in replications, such as sample characteristics and experimental interfaces, can significantly impact results (Huber et al., 2023; Holzmeister et al., 2024). Against this backdrop, I conducted a conceptual replication of GTMV's study, varying several contextual factors: data collection occurred during the COVID-19 pandemic, used different participant pools, and employed different experimental software with a modified user interface. This variation in implementation allows for testing the robustness of the original findings across different contexts.

Throughout this section, I pool data from both the citizens as well as the students from Hamburg and compare it with the students GMTV recruited in Maastricht. I refer to the two samples as the *replication sample* ($N_r = 116$) and the *original sample* ($N_o = 92$), respectively. For a more detailed comparison between student and non-student samples, see Appendix B. This three-way analysis between Maastricht students, Hamburg students, and Hamburg citizens further validates the robustness of our findings across populations.

3.2.1. Contribution Behavior

First, I ask whether the samples differ with respect to their initial contributions to the public good. Is the replication sample (consisting of both students and non-students) more pro-social than the original sample? A two-sided rank sum test reveals that it is not (p=0.393). Both samples contributed 10 tokens, that is, 50% of their endowments on average (median and mean). Moreover, both samples' initial contributions resemble the initial contributions participants usually make in the static game with partner matching.[4] However, in the dynamic game presented here, I am particularly interested in the subsequent periods because differences add up exponentially. Do the two groups remain similar over the course of time?

In particular, do the two samples' contributions follow the same path over the 10 periods they played? The answer is both *yes and no*. Figure 3 and Table 4 illustrate that the samples make similar contributions at the beginning and the end of the game but behave differently in between. More precisely, the left panel–depicting the average contributions in absolute terms–shows that the original sample contributed more than the replication sample *in all but the first and last period*.[5] For this reason, the original sample's behavior differs from the replication sample's behavior in two aspects: it contributes more and exhibits a considerable drop in the last period (whereas the replication sample's contributions flatten).

Note that increasing contributions over time imply increasing endowments over time. Hence, absolute contributions do not say much about the willingness to cooperate. For this reason, the right panel in Figure 3 shows the average *share of endowments contributed* over time. Both samples exhibit a similar

[4] See Figure 3B in Fehr and Gächter (2000, p. 989), for instance.

[5] In period five, this difference is significant: a two-sided rank sum test yields p=0.0486.

pattern: Their share of endowments contributed declined and did not stabilize. However, both samples also differ with respect to one aspect: the replication sample's share of contributions declines faster. This is mirrored by a two-sided rank sum test which is only significant at a five percent level in periods three and four.

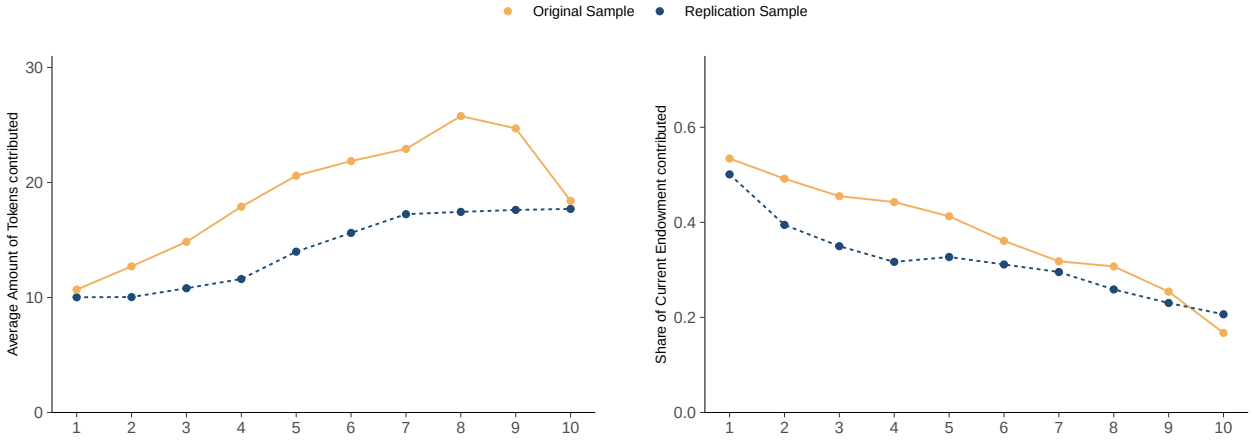


Figure 3: The average amount of tokens contributed over time across samples.

Again, both samples' behavior resembles the contributions participants usually make in the static game with partner matching: contributions equal approximately half of the endowments in the very first period and decrease to around ten percent of endowments by the last period.[6] In the dynamic game presented here, however, different paths lead to different levels of wealth – even if they share the same start- and endpoints. Next, we therefore turn our attention to the contributions' implications for wealth generation and growth.

3.2.2. Wealth Creation

How do the different contribution paths translate into wealth?[7] Given that the original sample contributed more, one would expect the respective groups to be more wealthy. A mere mean comparison indicates just that: An average group in the original sample accumulated about 478 tokens. In contrast, the average group in the replication sample accumulated about 380 tokens. This difference is insignificant at conventional levels though: A two-sided rank sum test (comparing differences between samples) yields p=0.136 for the mean stock in the last period of the game.

Although there clearly is growth, groups do not realize the maximal potential efficiency: under full cooperation, a group can accumulate a maximum of 4613 tokens or 230 Euros (24 times the minimum wage

[6]The right panel is thus, comparable to the visualizations *and results* in the static game. See, for instance, Figure 1B in Fehr and Gächter (2000, p. 986).

[7]To measure wealth and growth, I define a variable called *stock* which sums the endowments of all participants in a given group at the end of the round (that is, after the contributions have been made, multiplied, and redistributed).

at that time). This is depicted in the left panel of Figure 4, where one can see the average wealth over time by sample. The panel illustrates for both samples that growth was continuous and surprisingly linear, given the exponential character of the game's design. Despite somewhat differing contribution behavior between samples, neither the eventual wealth nor the corresponding growth paths differed. Differences in contribution behavior did, thus, not translate to significantly different wealth outcomes.

Why? Perhaps because the heterogeneity within samples and across groups has been too large to *detect* a significant difference. The right panel of Figure 4 depicts heterogeneity: In the replication sample, the richest group earned 1425 tokens (which is about 1781% of the initial endowment) whereas the poorest group ends up with 92 tokens (115%). More broadly, the replication sample is characterized by inequality across groups ($sd_{Replication} = 336.06$). The same holds true for the original sample ($sd_{Original} = 393.58$). Hence, the heterogeneity across groups does not differ between samples, which is remarkable because the replication sample was drawn from a more heterogeneous (non-convenience sample). Does it differ within groups?

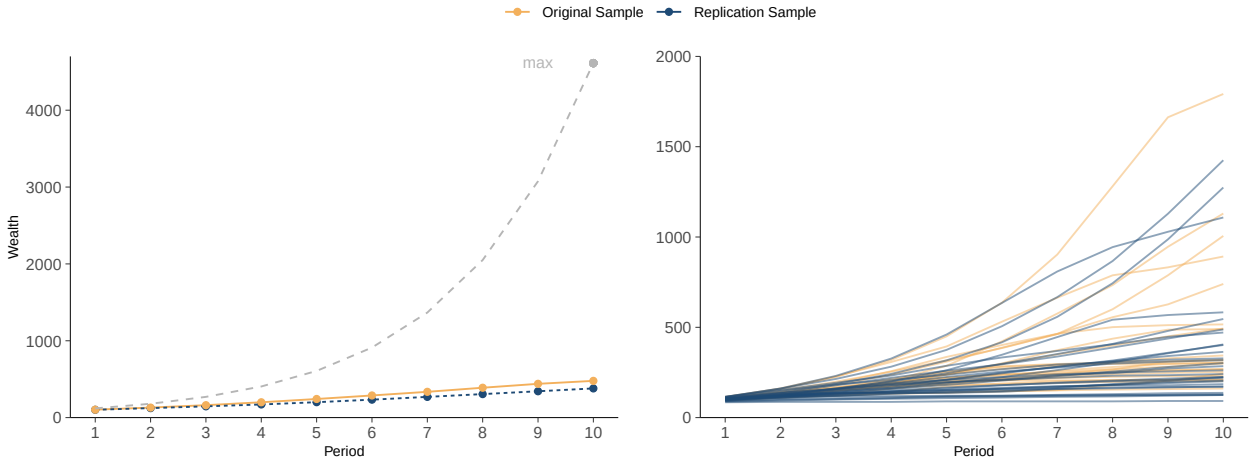


Figure 4: Average wealth over time across samples.

3.2.3. Inequality

Given the different samples and the possibility of endogenous growth—which essentially is the main feature of the game—I ask whether and how the inequality grows *within* groups. Figure 5 illustrates that inequality did grow: at the end of the game, the original and the replication groups exhibit an average Gini coefficient of 0.23 and 0.22, respectively.[8] Because every participant started with the same initial endowment (in *"period 0"*), every group started equally–with a Gini coefficient equaling zero.

[8]The two-sided rank sum test (comparing differences between samples) yields p=0.606 for the mean Gini coefficient in the last round of the game.

Figure 5 also shows that this initial state of equality ended with the first period already: both samples exhibit a stark incline in inequality before the second period started. From then on, the respective Gini coefficients grew slowly but continuously – for both samples.[9]

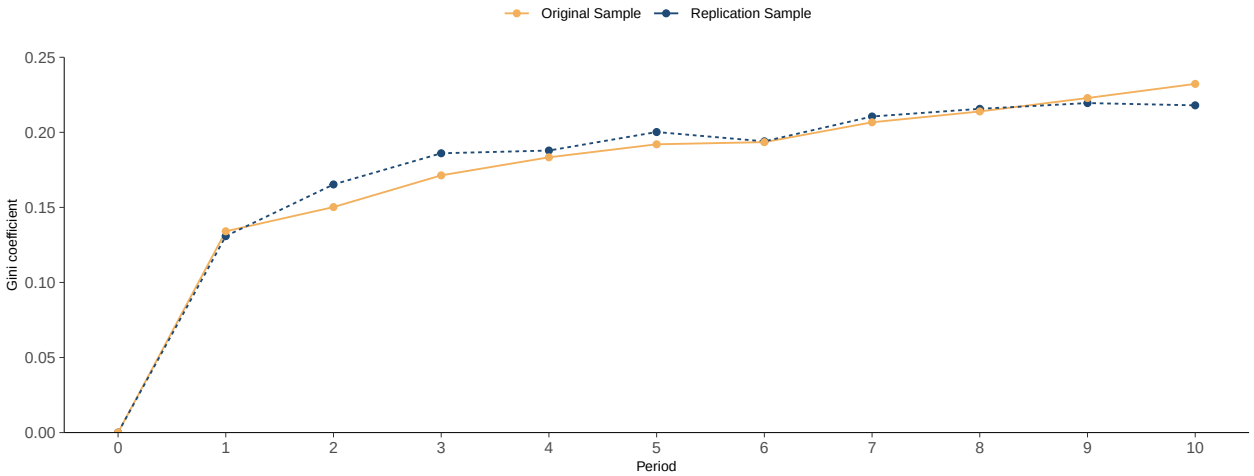


Figure 5: Average Gini coefficient (within groups) over time across samples

3.2.4. Discussion

In summary, we observe qualitatively similar patterns of contributions, wealth, and inequality–both within and across groups. We also find that early period wealth is strongly correlated with late period wealth. The Spearman correlation coefficient between periods 1 and 10 income is $\rho = 0.56$ ($p = 0.001$). In addition, the early period inequality is highly detrimental to final wealth, too (ρ = -0.38, $p = 0.04$).

Even though the sample sizes are small, these results suggest that the NOPUNISH 10 treatment is replicable and robust to different study environments (online instead of lab), alternative samples and time as the experiment was conducted during a global pandemic. Whereas one can consider this as one piece of evidence towards external validity, one may legitimately ask whether it also suggests generalizability. Hence, I'll examine whether the behavior observed in the experiment predicts real-world behavior next.

3.3. Generalizability

Whereas the replication results suggest robust behavior patterns in the abstract dPGG across different populations and time periods, an important question remains whether such behavior predicts real-world climate action. To examine this, I analyze the relationship between participants' choices in the dPGG and their subsequent VCA decisions to retire emission permits from the EU ETS.

[9] In each and every period, the two-sided rank sum test comparing gini coefficients between both sample yields p-values way over ten percent.

Focusing exclusively on the newly collected Hamburg samples, the following analysis proceeds in two steps: First, I examine the distribution of contributions across both tasks and subject pools. Second, I use a censored regression to test whether dPGG behavior systematically predicts VCA decisions, controlling for demographic characteristics.

3.3.1. VCA Predictability Across Samples

Figure 6 shows kernel density distributions across four panels, comparing VCA and first-period dPGG contributions separately for students and non-students. In both samples, VCA distributions (top panels) are right-skewed with a mode at zero, indicating that a significant portion of participants chose not to contribute to climate mitigation. First-period dPGG contributions (bottom panels) show higher averages and fewer zero contributions, suggesting participants were more willing to cooperate in the abstract game setting.

Figure 7 visualizes the relationship between tasks through two scatter plots, separated by sample group. Both panels include a fitted line with confidence intervals. The scattered pattern and nearly flat fitted lines indicate no systematic relationship between dPGG and VCA contributions in either sample, suggesting that cooperative behavior in the dynamic game setting may be driven by different motivational factors than real-world climate action decisions.

To formally test for relationships between behavior in both tasks while accounting for demographic factors, I estimate Tobit regression models with VCA donations as the dependent variable (censored at 0 and 100 percent of earnings) in Table 8 (see Appendix C). The baseline specification shows no significant correlation between first-period dPGG contributions and VCA donations (Model 1). Student status emerges as a significant predictor in Model 2 (β = -48.68, $p < 0.05$), this effect becomes insignificant when controlling for age in Model 3. However, given the minimal age overlap between students and non-students shown in Figure 1, the age coefficient ($\beta = 0.92$, $p < 0.05$) likely captures the same between-group differences rather than true age effects or vice versa. Most importantly, across all specifications, there remains no significant correlation between dPGG contributions and VCA decisions.

The absence of correlation contrasts with expectations that the dPGG's more ecologically valid, dynamic structure–which captures temporal interdependencies characteristic of climate change–would make it a better predictor of real climate action compared to standard public goods games. However, as both Figure 7 and Table 8 document, there is substantial noise in the relationship between contributions across the two tasks, suggesting that different motivational factors may drive behavior in each context.

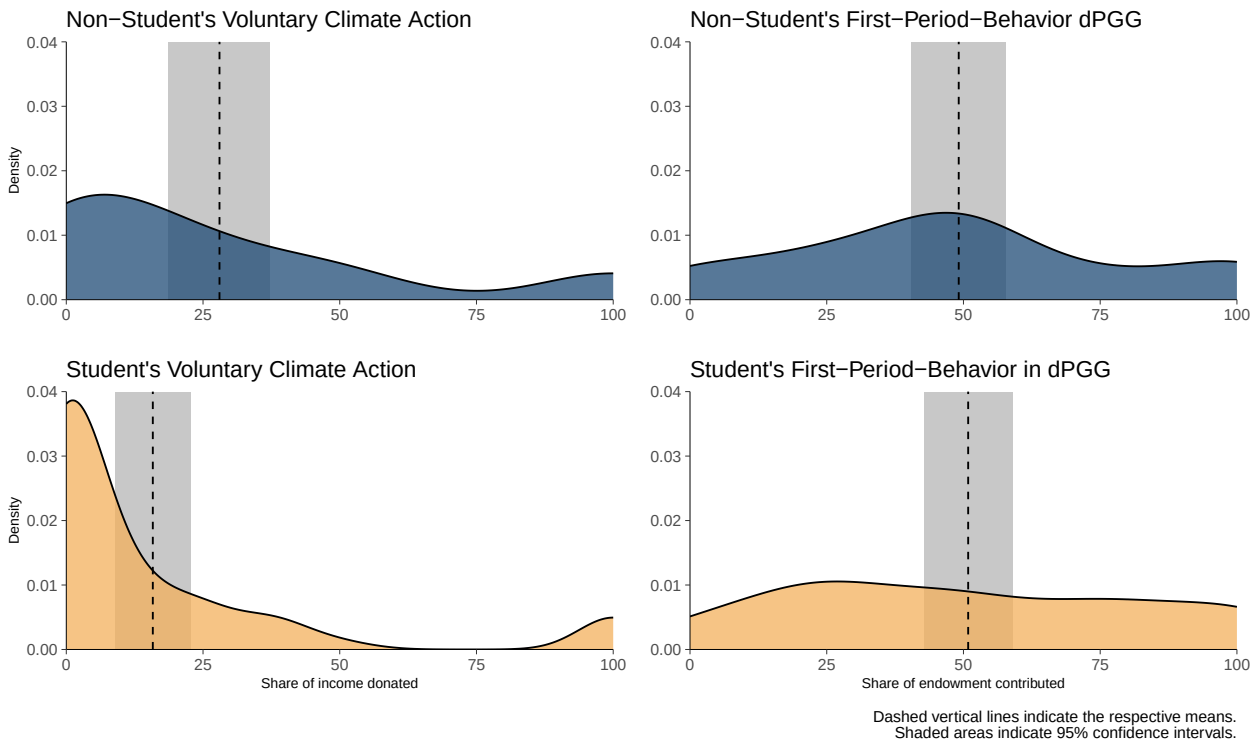


Figure 6: Kernel distributions of contributions across tasks and subject pools.

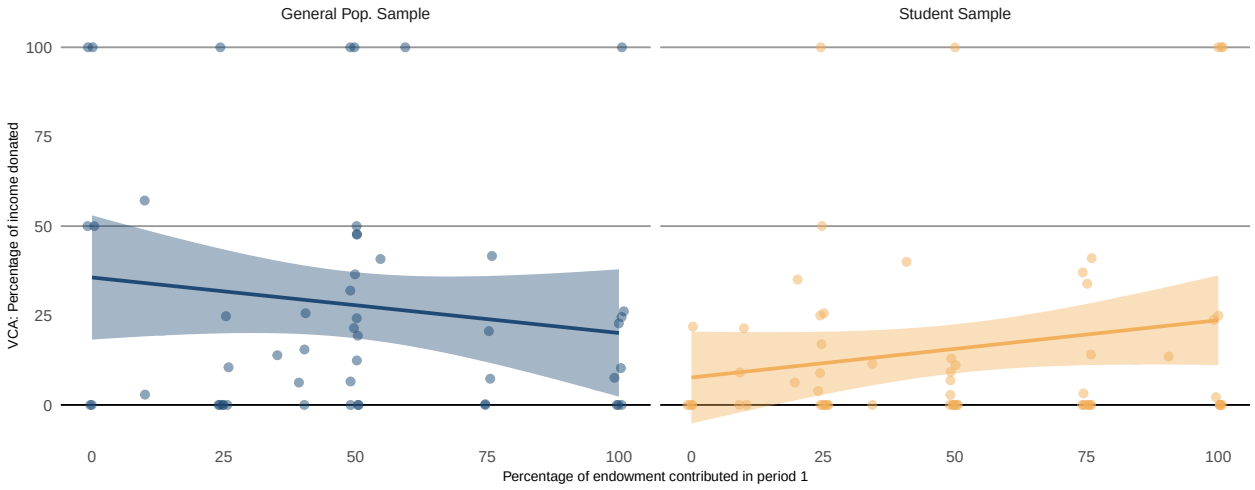


Figure 7: Scatter plot of average contributions in the dPGG and real giving task.

3.3.2. Discussion

The findings align with Goeschl et al. (2020)'s conclusion that abstract public goods games have limited predictive power for real-world climate action, despite incorporating relevant structural features like temporal interdependence. This limitation persists across both student and the more representative sample, suggesting the challenge lies beyond subject pool characteristics. Acknowledging that sample sizes are small and absence of evidence does not imply evidence of absence, the results indicate that while the dynamic game offers higher ecological validity than static games for climate contexts, experimental paradigms face inherent constraints: participants make decisions in artificial laboratory sessions with compressed time frames, under researcher scrutiny, and following detailed instructions. These fundamental differences between experimental and natural decision environments suggest that laboratory games may be better suited for testing specific behavioral mechanisms than for predicting real-world climate actions.

4. Conclusion

This paper confirms that GMTV's findings are both reproducible and replicable. Despite minor computational issues in the original analysis, the qualitative patterns remain robust: contributions decline as a share of endowments but increase in absolute terms, generating substantial wealth inequality both within and across groups. These patterns persist across different populations (students and general public) and experimental implementations (laboratory and online), suggesting the findings capture fundamental features of dynamic social dilemmas.

But do these experimental results translate to real-world behavior? The evidence presented here suggests they may not. Despite the dynamic public goods game incorporating temporal interdependencies—a key feature of climate change—behavior in the abstract game shows no meaningful correlation with actual climate action through voluntary carbon offsetting. This disconnect persists across both student and more representative samples, suggesting the gap between experimental and field behavior extends beyond subject pool effects.

Rather than attempting to predict specific behaviors, dynamic public goods games may be more valuable for isolating and understanding particular behavioral mechanisms. The strong replicability of the results suggests the paradigm reliably captures how people navigate trade-offs between individual and collective interests when current choices shape future possibilities. This mechanistic insight, while perhaps not directly predictive, can still inform policy design by highlighting how institutional features might enable or constrain cooperative solutions to climate challenges.

The disconnect between experimental and field behavior likely stems from fundamental differences in decision environments. While the dynamic structure enhances the game's ecological validity through temporal

interdependencies, participants still make choices in compressed time frames, under researcher scrutiny, and following detailed instructions—conditions far removed from real-world climate decisions.

Several limitations of this study warrant mention. First, the sample sizes—particularly in the more representative sample—are relatively small, limiting statistical power. Second, the voluntary climate action measure captures only one specific type of climate-related behavior; other forms of climate action might show different patterns. Third, the online implementation during the COVID-19 pandemic may have influenced participants' decision-making in ways that differ from traditional laboratory settings. Despite these limitations, the robust replication of GMTV's core findings across different populations and implementation methods suggests the paradigm captures fundamental aspects of dynamic social dilemmas, even if these don't directly translate to field behavior.

Open Science Disclaimer

In the interest of transparency, it is important to note that the analyses in Section 3.3 deviate from the pre-registered research plan[10] in two ways. First, the recruitment of student participants was not initially planned but became necessary when the more representative sample pool was exhausted. Second, while the comparison between abstract game behavior and voluntary climate action was part of the original design, the detailed analysis of generalizability presented here extends beyond the pre-registered scope. These deviations emerged organically during the research process.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work I used *Claude* (Anthropic) in order to assist with language editing and grammar revisions. After using this tool/service, I reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix

A: Recreation Reproduction

I identified two computational errors and one conceptual issue in the original data[11]. The authors promptly addressed these issues and provided helpful clarifications. Importantly, the main results of the paper remain robust to these corrections.

Error 1: The Gini coefficient

The Gini coefficient calculation contains an error when two group members have identical endowments, as the program fails to rank these members for subsequent calculations. Table 1 illustrates this for group 101 in period 5: While the Gini coefficient should equal `GINI`=0.127 for all group members, participants `112` and `113`, who have equal endowments, deviate from this value. The `DescTools::Gini()` function in R handles these ties correctly and was therefore used for all calculations.

Table 1: Subset of Data illustrating the Gini Coefficient's Error

exp_num	gr_id	per	subj_id	tokens	other1	other2	other3	gini	GINI
1	101	5	111	42	27	27	30	0.127	0.127
1	101	5	112	27	42	27	30	0.111	0.127
1	101	5	113	27	42	27	30	0.111	0.127
1	101	5	114	30	42	27	27	0.127	0.127

Error 2: The share of endowments contributed

The original analysis incorrectly calculates the share of endowments contributed (`mean`) by using a lagged endowment measure (`gdp`). This error stems from the STATA code:

```
*tsset subj_id per
*gen mean=sum/l.gdp
```

Table 2 illustrates the issue by showing data from participant `111` in group `101` over three periods. The `sum` column shows the group's total contributions (e.g., 18 tokens in period 4), while `gdp` shows the group's total endowment at the start of each period (e.g., 116 tokens in period 4). The original calculation (`mean`) incorrectly divides each period's contributions by the *previous* period's endowment. For example, in period 5, the authors calculated $\frac{18}{116} = 0.155$ instead of $\frac{18}{126} = 0.143$. As shown in the table, calculating `MEAN=sum/gdp` solves this error.

[11]The pre-processed data can be found in the supplementary materials they provide in their online appendix.

Table 2: Subset of Data illustrating the Means's Error

exp_num	gr_id	per	subj_id	gdp	sum	mean	MEAN
1	101	4	111	116	18	0.168	0.155
1	101	5	111	126	18	0.155	0.143
1	101	6	111	136	17	0.135	0.125

The misconception: Timing

The authors note that the Gini coefficient and wealth measures refer to the situation at the start of a period, clarifying that wealth is defined as the endowment at the beginning of the following period. This definition switched during paper revisions.

I argue for measuring wealth at the end rather than the beginning of periods for two reasons: First, beginning-of-period wealth is redundant with the existing endowment measure. Second, beginning-of-period measures miss crucial end-game behavior. For instance, the very last value would only capture the beginning of the final period, missing the often significant changes in contributions during that period. Moreover, if wealth in period 1 is defined as the endowment at the beginning of period 1, it would be four times the initial endowment by design, without capturing any behavioral outcomes.

This timing choice also affects Gini coefficient calculations: While inequality is zero by design before any decisions are made (as all participants receive equal endowments), we cannot observe final inequality under the current definition. Therefore, I define wealth and inequality measures as end-of-period outcomes throughout my analysis.[12]

[12]For consistency with the original paper's presentation in Table 1, I show the Gini coefficients as they were originally calculated. However, these values represent the inequality level at the beginning of the current period (or equivalently, the end of the previous period), not the definition I use in my analysis which measures inequality at the end of the current period.

B: Comparison Across Population Samples

This appendix extends the main analysis by separating the Hamburg sample into student and non-student subgroups, enabling three-way comparison with the original GMTV Maastricht student sample. Figures 8-10 track contribution patterns, wealth accumulation, and inequality development across all three populations. The consistent patterns across both student samples (Hamburg and Maastricht) validate the original replication findings, while the similarities with the more representative sample suggest broader generalizability beyond student populations.

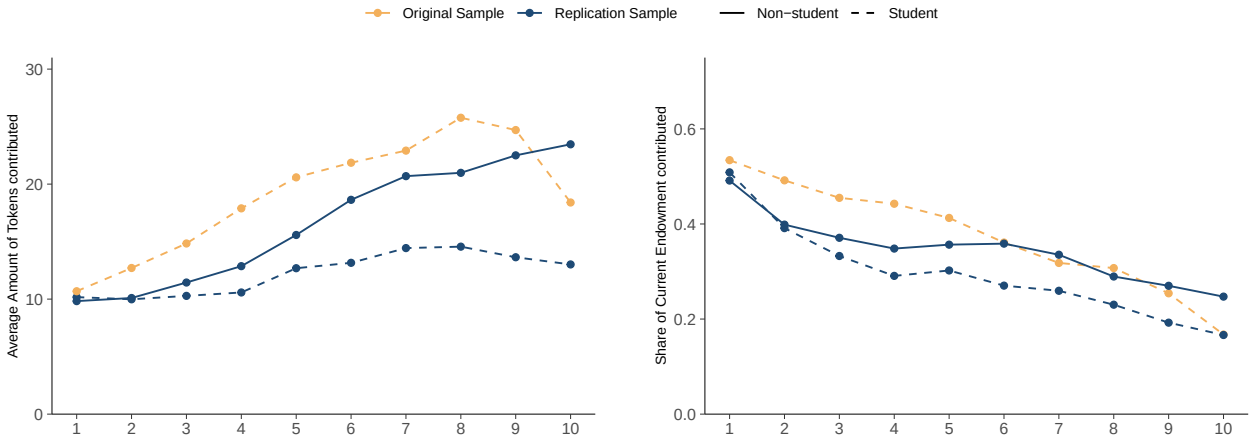


Figure 8: The average amount of tokens contributed over time across samples.

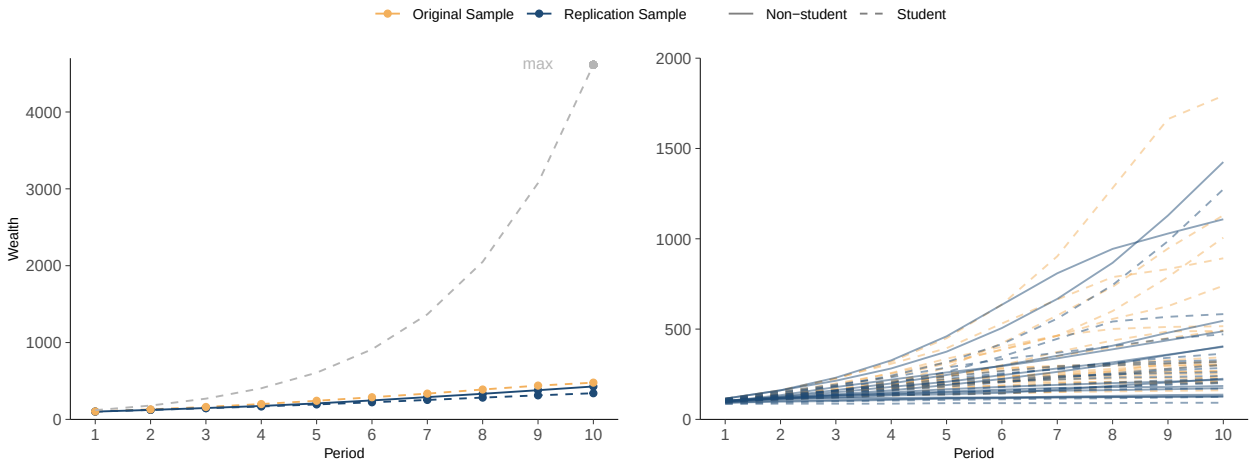


Figure 9: Average wealth over time across samples.

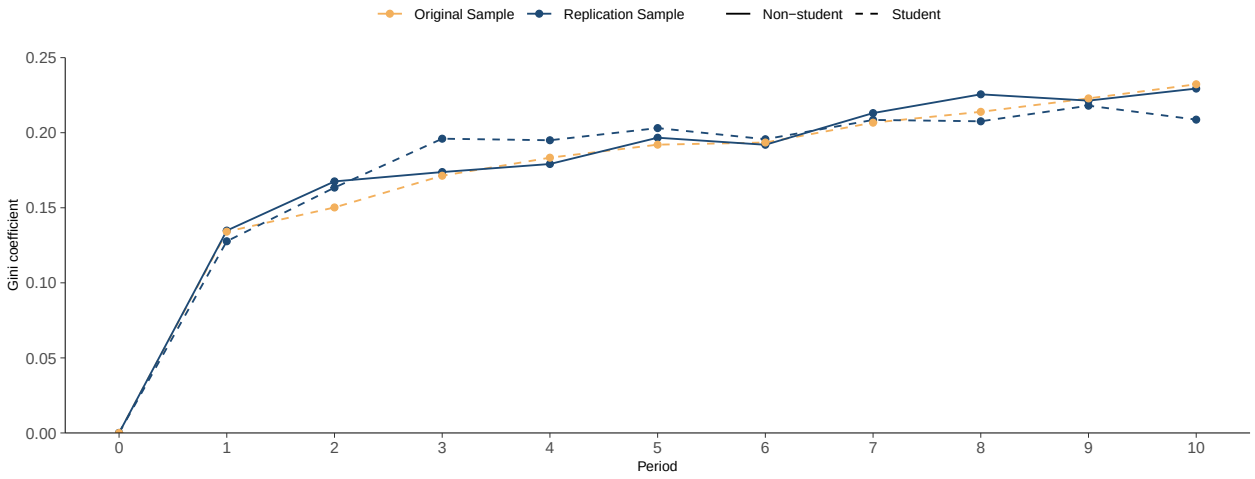


Figure 10: Average Gini coefficient (within groups) over time across samples

C: Tables

This appendix presents the data shown in Section 2, 3 and 4's figures in tabular form. Table 3 reports sample porpoerties whereas Tables 4-7 compare outcomes between the original GMTV sample and our replication sample, reporting means, confidence intervals, and statistical tests for contributions, wealth, and inequality measures. Table 8 presents a Tobit regression discuessed in Section 3.3.

Table 3: Sample Properties

Sample	Hamburg Citizens			Hamburg Students			Maastricht Students			
Variable	N	Mean	SD	N	Mean	SD	N	Mean	SD	Test
Age	52	48	15	64	26	7.6	92	35	2.6	F= 86.246***
Female	52	0.48	0.5	64	0.56	0.5	92	0.38	0.49	F= 2.596*

Statistical significance markers: * p<0.1; ** p<0.05; *** p<0.01

Table 4: The average amount of tokens contributed over time across samples.

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
1	42.739	38.367 - 47.111	40.069	34.406 - 45.732	0.561
2	50.826	42.775 - 58.877	40.138	31.557 - 48.719	0.061
3	59.348	45.916 - 72.780	43.207	30.815 - 55.599	0.067
4	71.565	53.030 - 90.100	46.414	29.930 - 62.898	0.014

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
5	82.348	55.398 - 109.298	55.931	33.101 - 78.761	0.049
6	87.435	45.735 - 129.135	62.448	31.121 - 93.776	0.191
7	91.652	36.859 - 146.446	68.966	33.626 - 104.305	0.740
8	103.087	30.382 - 175.792	69.759	29.621 - 109.896	0.507
9	98.804	20.125 - 177.483	70.448	22.055 - 118.841	0.747
10	73.609	20.019 - 127.198	70.793	15.116 - 126.470	0.606

Table 5: The average share of current endowments contributed over time across samples.

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
1	0.534	0.480 - 0.589	0.501	0.430 - 0.572	0.561
2	0.487	0.418 - 0.557	0.383	0.309 - 0.457	0.051
3	0.441	0.358 - 0.524	0.325	0.245 - 0.406	0.046
4	0.418	0.338 - 0.497	0.278	0.202 - 0.354	0.009
5	0.377	0.285 - 0.470	0.270	0.190 - 0.349	0.063
6	0.300	0.194 - 0.407	0.234	0.153 - 0.315	0.277
7	0.240	0.143 - 0.337	0.217	0.150 - 0.285	0.934
8	0.214	0.122 - 0.306	0.176	0.111 - 0.240	0.593
9	0.165	0.081 - 0.249	0.149	0.087 - 0.212	0.949
10	0.115	0.049 - 0.181	0.121	0.067 - 0.175	0.450

Table 6: The average wealth over time across samples.

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
1	103.348	100.966 - 105.729	101.724	98.806 - 104.642	0.605
2	130.435	124.592 - 136.277	123.517	116.597 - 130.437	0.099
3	161.696	149.708 - 173.683	146.517	133.780 - 159.255	0.061
4	199.522	178.632 - 220.412	171.000	150.455 - 191.545	0.033
5	242.217	208.842 - 275.593	200.586	169.020 - 232.152	0.033
6	288.174	235.294 - 341.054	233.448	186.751 - 280.146	0.047

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
7	335.478	256.590 - 414.367	269.586	205.759 - 333.413	0.072
8	388.739	275.392 - 502.086	305.897	223.488 - 388.305	0.107
9	439.674	290.308 - 589.040	342.759	239.624 - 445.893	0.124
10	478.087	307.892 - 648.282	379.828	251.997 - 507.658	0.136

Table 7: The average Gini coefficient over time across samples.

Period	Mean: Original	95%-CI: Original	Mean: Reproduction	95%-CI: Reproduction	p-Value
1	0.134	0.111 - 0.157	0.131	0.111 - 0.151	0.818
2	0.150	0.118 - 0.182	0.165	0.139 - 0.192	0.733
3	0.171	0.133 - 0.210	0.186	0.153 - 0.219	0.495
4	0.183	0.137 - 0.230	0.188	0.150 - 0.226	0.768
5	0.192	0.141 - 0.243	0.200	0.159 - 0.241	0.606
6	0.193	0.139 - 0.248	0.194	0.149 - 0.239	0.825
7	0.207	0.153 - 0.260	0.211	0.165 - 0.256	0.905
8	0.214	0.160 - 0.268	0.216	0.170 - 0.261	0.985
9	0.223	0.171 - 0.275	0.220	0.173 - 0.266	0.726
10	0.232	0.179 - 0.285	0.218	0.171 - 0.265	0.606

D: Online Feasibility

Throughout this section, I do not consider GMTVs' data and pool data I gathered from the citizens as well as the students from Hamburg (N = 151).

How did the participants, who have never participated in an online group experiment before, cope with the situation? Moreover, did participants understand the unfamiliar setting they found themselves in? While the answer to the former question requires more thought, the answer to the latter simply is *yes*: 67 out of 116 answered with *"yes"* when I asked them. Another 44 answered with *"rather yes"* while nobody indicated that he or she did not understand the situation at all. Some behavioral data are supporting this finding: The user interface offered a popup to review instructions or contact information. I tracked both and find that none of the participants ever opened these popups even though they were clearly visible in the decision screens' header and introduced in the instructions. To further analyze how participants coped with the

Table 8: Correlations between first-round-dPGG and VCA behavior

	Tobit Regressions		
	VCA as Share of Earnings		
	(1)	(2)	(3)
First-period contribution in percent	0.07 (0.18)	−0.17 (0.25)	−0.23 (0.25)
Student (1 = yes)		−48.68** (20.44)	−31.06 (21.74)
First-period contr. x Student		0.46 (0.34)	0.50 (0.34)
Age			0.92** (0.45)
Female (1 = yes)			4.98 (10.74)
logSigma	4.00*** (0.11)	3.96*** (0.11)	3.94*** (0.11)
Constant	3.16 (10.70)	29.35** (14.67)	−13.93 (26.41)
Observations	116	116	116
Log Likelihood	−359.78	−356.07	−354.01
Akaike Inf. Crit.	725.55	722.15	722.02
Bayesian Inf. Crit.	733.82	735.91	741.29

Note: *p<0.1; **p<0.05; ***p<0.01

situation, I consider three additional metrics: selection into the experiment, attrition as well as the time spent on each page.

I first comment on the selection into the experiment: It was difficult to recruit the sample. The panel counted 1.209 non-students of which I was able to recruit 130 participants who finished the experiment—even though I varied the weekdays and timing of the sessions (which were conducted during a nationwide lockdown under mandatory home office regime). For this reason, I also recruited students in the last session which explains the relatively large number of showups in Table 9. Although I intended to refrain from the recruitment of students initially, this particular sub-sample enabled me to investigate the generalizability of my results as I discuss in Section 3.3.

Table 9: The Experimental Sessions' Meta Data

Session Code	Date	Time	Showups	Dropouts	Residuals	Participants	Observations
jyf8xd0s	2021-07-01	15:00	35	4	3	28	7
vggk2gh1	2021-07-03	13:00	20	8	0	12	3
8gi7c8xg	2021-07-09	13:00	21	5	4	12	3
d6jrsxnr	2021-07-23	14:00	75	8	3	64	16

Turning to the time spent on each page, I focus on the decision times in the dynamic public goods game

as Anderhub et al. (2001) did. How many seconds did the participants need to decide in each period of the game? Not too many. Figure 11 illustrates an plausible pattern: The first decision took about 22 seconds. The second decision—where participants first learned about the other group members' previous decisions—took longer (about 33 seconds). Subsequently, decision times first declined and stabilized at 19 seconds. Importantly, decision times were so short that crosstalk, that is, communication through private channels—a common concern[13] in online experiments—was unlikely, especially because it would require the identification of other group members.[14]

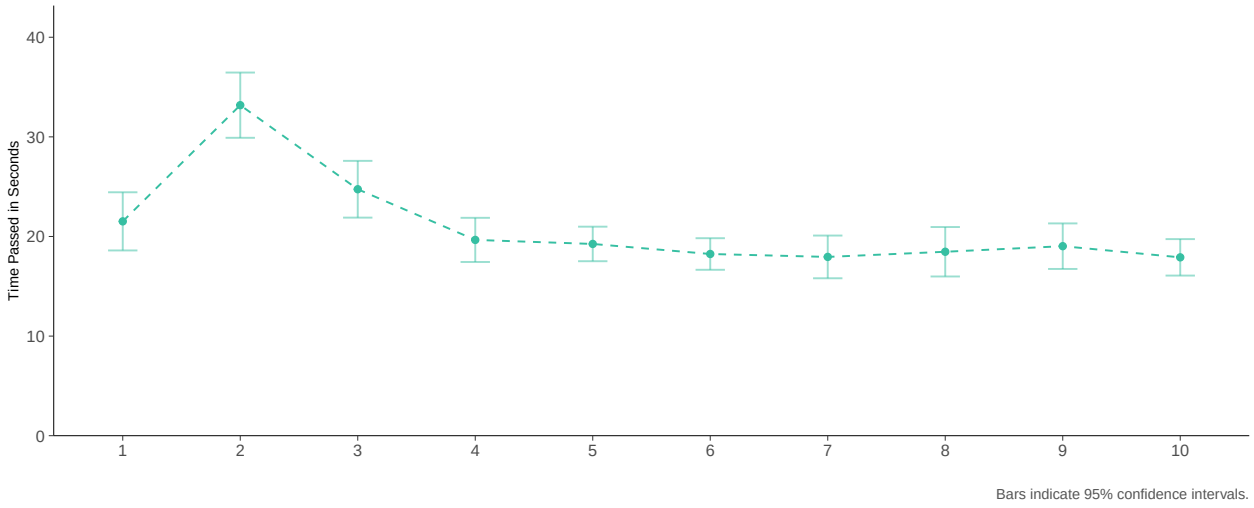


Figure 11: Average Time Spent for Each Contribution per Period

Considering attrition, I find that it did not affect the interactive experiment at all. To elaborate, I differentiate between dropouts and residuals: Participants who could not be matched to other group members are called residuals. Participants who intentionally left the experiment are called dropouts. Residuals did not participate in the experiment *by design*. Dropouts did not participate in the experiment *by choice*. Out of 151 people who showed up, I count 10 residuals and 25 dropouts. All of the residuals waited to be matched to a group unsuccessfully before they got paid one Euro for their patience. In contrast, all of the dropouts left while reading the instructions and before being matched to other group members. Moreover, they received no payment at all. Hence, attrition was no concern considering the dynamic public goods game or the expenses.

[13] See, for instance, the discussion section in Arechar et al. (2018, p. 119).

[14] There were only 9 participants (from all four sessions) who needed more than 60 seconds to make the second decision.

E: Comprehension

The participants in this study were grouped based on the speed at which they answered comprehension questions, with higher group IDs indicating that those participants completed the questions later. This could introduce a potential bias in how participants were grouped in terms of their comprehension levels. Specifically, participants who were quicker to answer comprehension questions may have had a better understanding of the task, and thus their comprehension scores could be systematically higher than those who took longer to answer. This grouping by answering speed might inadvertently align with comprehension ability, potentially leading to confounding effects in the results.

To test this potential bias, I estimate a linear mixed-effects model specified as:

$$y_{ij} = \beta_0 + \beta_1 \text{GroupID}_{ij} + u_j + \epsilon_{ij}$$

where y_{ij} represents the comprehension score for participant i in session j, GroupIDij indicates their grouping order, u_j captures session-specific random effects, and ϵ_{ij} is the error term. The model tests whether faster completion of comprehension questions predicts better understanding while accounting for potential session-level variations through random intercepts.

The results presented in Table 10 suggest that group ID does not significantly predict comprehension. The coefficient for group ID is small and insignificant (β = -0.02, se = 0.01), indicating that the order in which participants were grouped based on their response speed does not have a significant effect on their comprehension scores in this sample.

Table 10: Linear Mixed-Effects Model for Comprehension, Accounting for Group ID and Session Effects

	Mixed Effects Model
	Comprhension
Group ID	−0.02 (0.01)
Constant	2.65*** (0.16)
Observations	116
Log Likelihood	−101.69
Akaike Inf. Crit.	211.38
Bayesian Inf. Crit.	222.39
Note:	*p<0.1; **p<0.05; ***p<0.01